GridIQ

Site assessment: overview

Demo Corporation · Headquarters

PROVISIONAL: the assessment is still running. Results cover the elapsed period only and are not final.

Assessment
GQ-A-0002 · 30-day assessment

Report
revision 0 · provisional preview

Observation period (SAST)
2026-10-22 21:00 -> 2026-11-12 07:26

Observed
20.43 days of 30

Site address
123 Power Lane, Johannesburg

Measurement points
Phase1 Test Meter (incomer), Main Incomer (incomer)

Prepared by GridIQ from measured data. No engineer's sign-off is implied unless a named reviewer appears on this page.

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Executive assessment

Main measurement point: Main Incomer. Period 2026-10-22 21:00 to 2026-11-12 07:26 SAST.

On weekdays the load is highest between 09:00 and 17:00 (median of 10 weekdays).

Outside operating hours the load falls to 26 % of its operating-hours level.

1 area merits follow-up.

Key numbers

Energy imported
24 431 kWh

Maximum demand
190.5 kVA

Load factor
27 %

Areas to follow up
1 of 4

Energy imported: Register energy over the observed 490 h (mixed). Maximum demand: Highest 30-minute average (billing basis); contracted 300 kVA. Load factor: Average kW ÷ maximum 30-minute kW. Typical operating load 152 kW. Areas to follow up: 1 prioritised observation(s); 0 area(s) with insufficient evidence.

Assessment by area

Area	Status	Basis
Measurement quality	Investigate	63 % of expected 10-minute slots measured (rule: investigate below 80 %).
Supply observations	No material issue observed	Voltage, frequency and voltage THD within the NRS 048-2 screening thresholds in the observed period.
Installation and load behaviour	No material issue observed	Loaded PF 0.968, current unbalance P95 – %, after-hours load 26 % of operating.
Demand exposure	No material issue observed	Maximum 30-minute demand 190.5 kVA, 64 % of the contracted 300 kVA.

Immediate priorities

- 1. Only 63 % of the expected measurements were received; conclusions for this point are limited. Next: Restore the meter connection; GridIQ will check what data can be recovered. (GridIQ; high confidence)

Important measurement limits

- Observation period 20.4 days (2026-10-22 21:00 to 2026-11-12 07:26 SAST); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.

1. Findings by point

- Investigate (Phase1 Test Meter, confidence high; for: GridIQ): Only 0 % of the expected measurements were received; conclusions for this point are limited.
- Investigate (Main Incomer, confidence high; for: GridIQ): Only 63 % of the expected measurements were received; conclusions for this point are limited.

2. Installation and measurement boundary

Point	Role	Nominal	CT primary	Contracted MDI	Device · installed (SAST)
Phase1 Test Meter	incomer	230 V	–	300 kVA	device · 2026-11-03 20:52
Main Incomer	incomer	230 V	–	300 kVA	device · 2026-07-26 22:50

Point	Coverage	10-min slots measured	Samples	Reporting interval	Gaps >= 30 min
Phase1 Test Meter	0.0 %	1 of 2 942	3	32 s	2
Main Incomer	63.4 %	1 866 of 2 942	3 732	300 s	1

Coverage: unique 10-minute slots of the observation window; each slot counts min(1, samples × its own reporting interval ÷ 600 s); samples are unique per meter and timestamp (duplicates are rejected at ingest). Missing data is never filled or shown as zero.

sum of 2 top-level incomer(s); sub-meters are inside these and are not added

3. Energy, demand and operating pattern

Phase1 Test Meter

Energy imported
1,7 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
–

Contracted MDI
300 kVA (–)

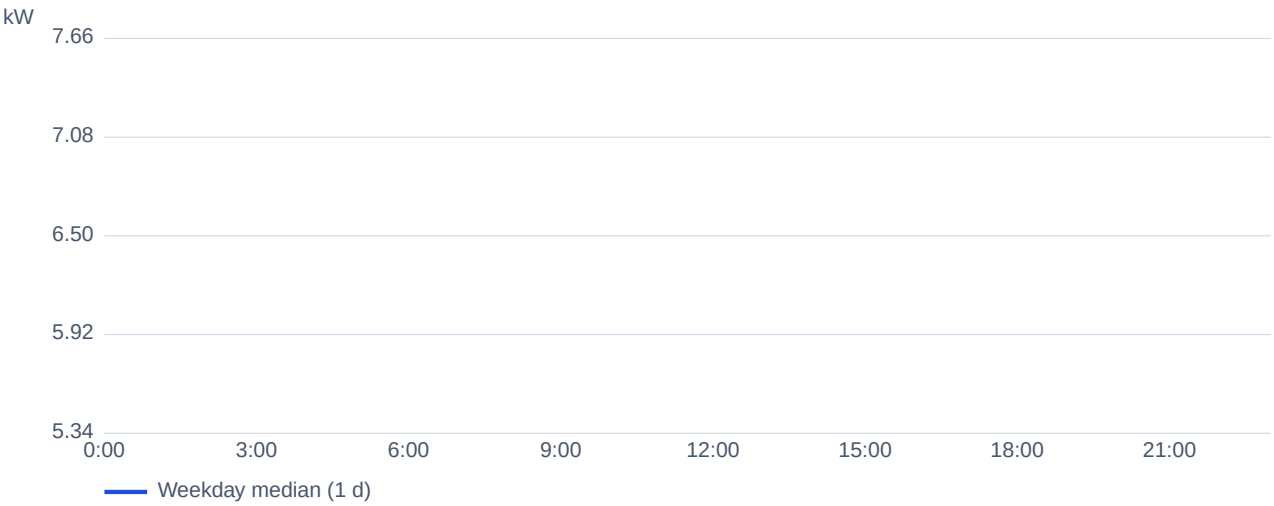
Highest instantaneous active power
7,2 kW (minute of 2026-11-03 20:53)

Baseload (10th percentile)
6,5 kW

Operating-hours average
–

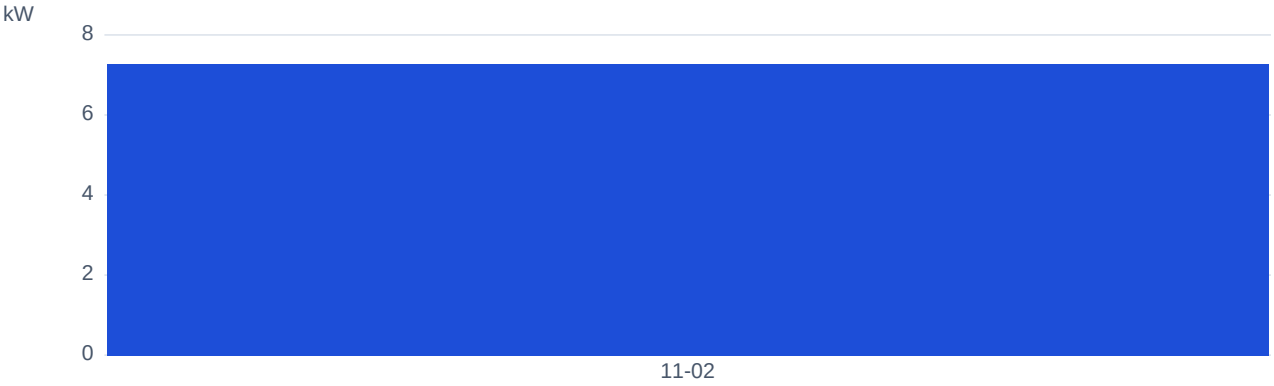
Outside operating hours
6,5 kW

Figure 1. Operating fingerprint: typical day (hourly kW, SAST)



hourly average kW, SAST; Monday to Friday; median with P10–P90 across days when at least 3 days contribute to an hour

Figure 2. Daily maximum active power (SAST days)



hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

Main Incomer

Energy imported
24 430,8 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
190,5 kVA at 2026-11-03 14:00

Contracted MDI
300 kVA (64 % used)

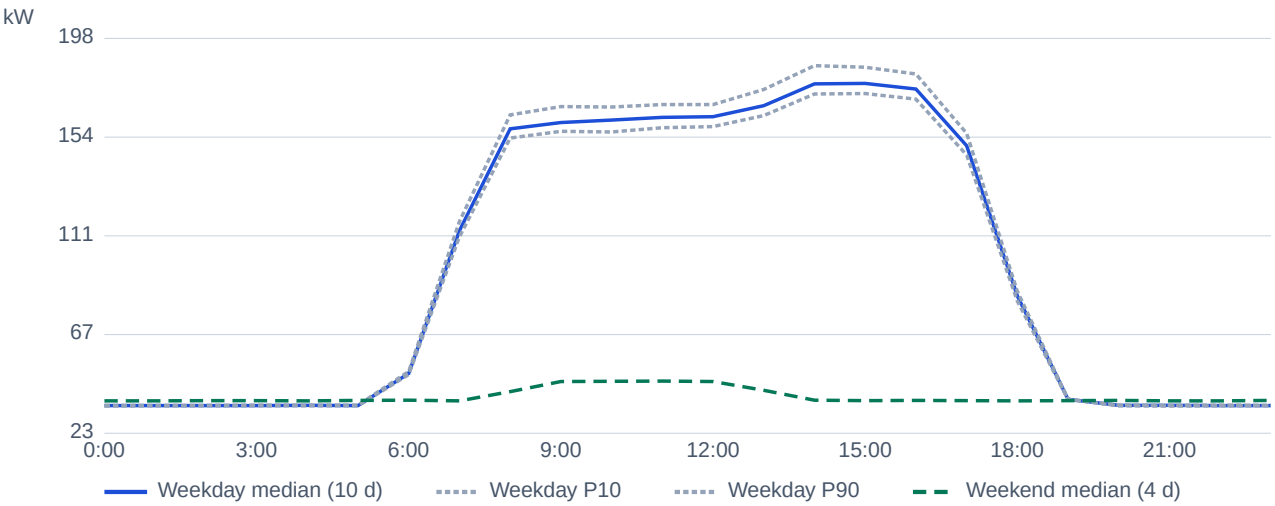
Highest instantaneous active power
192,4 kW (minute of 2026-11-03 14:00)

Baseload (10th percentile)
34,8 kW

Operating-hours average
152 kW

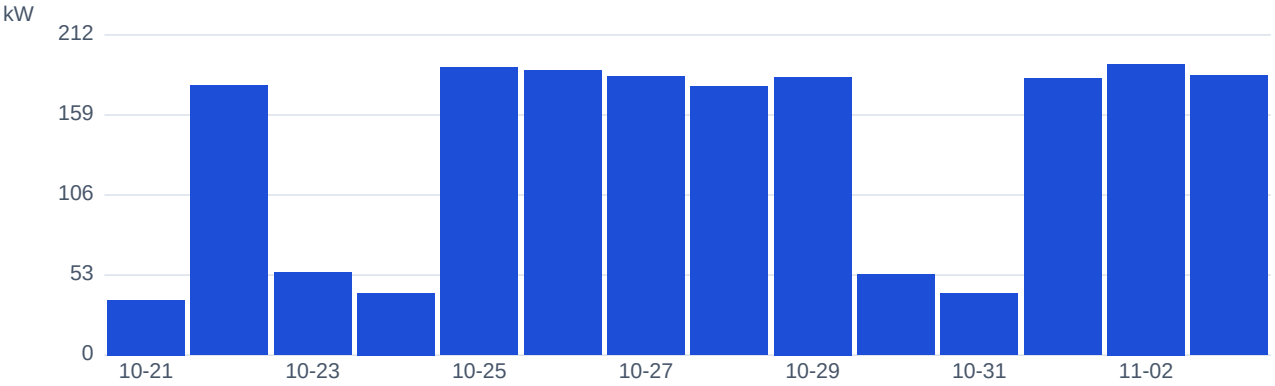
Outside operating hours
39,1 kW

Figure 3. Operating fingerprint: typical day (hourly kW, SAST)



hourly average kW, SAST; Monday to Friday; median with P10–P90 across days when at least 3 days contribute to an hour

Figure 4. Daily maximum active power (SAST days)



hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

4. Electrical summary

- Phase1 Test Meter: voltage A 229,9–231,5 V, B 229,8–229,8 V, C 231–231 V; THD-V p95 not measured; PF (loaded) 0.957; 0 minute(s) below 23 V.
- Main Incomer: voltage A 226–234,7 V, B 225,7–233,9 V, C 226,6–235,6 V; THD-V p95 A 2.7 %; PF (loaded) 0.968; 0 minute(s) below 23 V.

Screening against NRS 048-2 thresholds; see the power-quality report for detail and the basis of every figure.

5. Notable events and data limitations

- PROVISIONAL: the assessment is still running; these results cover the elapsed period only and are not final.
- Observation period 20.4 days (2026-10-22 21:00 to 2026-11-12 07:26 SAST); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.

- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.
- Phase1 Test Meter: 0 % of expected measurements received (2 gap(s) of 30 min or more).
- Phase1 Test Meter: the meter's capability profile is not confirmed; only channels present in the data are reported.
- Main Incomer: 63 % of expected measurements received (1 gap(s) of 30 min or more).
- Main Incomer: the meter's capability profile is not confirmed; only channels present in the data are reported.

0 voltage/frequency excursion event(s) recorded; inferred from the sampled RMS trend (pq_events), not device-recorded.

6. Prioritised actions

#	Action	Who investigates	Urgency	Confidence
1	Only 0 % of the expected measurements were received; conclusions for this point are limited.	GridIQ	next maintenance	high
2	Only 63 % of the expected measurements were received; conclusions for this point are limited.	GridIQ	next maintenance	high

7. Feasibility-data readiness

Supplied: 30-minute interval load profile (kW, kVA); daily energy; billed maximum demand with timestamps; power-quality screening results; metering topology and observation limits.

Missing inputs: declared operating hours (default 06:00–18:00 weekdays used); supply details and tariff; existing PV / generator / battery details; fault level at the point of supply; a full month or more of data (seasonal and monthly patterns).

GridIQ supplies measured data; system sizing and financial modelling belong to the feasibility study.

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
3af4be60694ccb426525bf76ab2cd3ab…

Calculation version
gridiq-calc-2026.11.2

Rule version
gridiq-rules-2026.11.2

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-12 07:26

Generated (SAST)
2026-11-12 07:26

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 3af4be60694ccb426525bf76ab2cd3abda25fda79199f2ee93e43bb6c9dcf688